Yewon Joung

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Research Interests

Machine Learning for Healthcare, Natural Language Processing, Transfer Learning

Education

University of North Carolina at Chapel Hill — *Chapel Hill, NC, United States*
M.S. in Computer Science — Aug. 2025 – Present

Sookmyung Women's University — *Seoul, South Korea*
B.E. in Mechanical Engineering — Mar. 2019 – Feb. 2024
- GPA: 4.25/4.5 (96.99/100)

Research Experience

Autonomous Mechanical Systems Laboratory, Sookmyung Women's University — *Seoul, South Korea*
Research Intern (Advisor: Prof. Jooyong Sim) — Dec. 2020 – Sep. 2022

- **Multimodal Depression Detection Using Psychiatric Consultation Data**:
  Developed a multimodal depression detection model (Bi-LSTM + Spectral CNN) using Korean psychiatric consultation data provided by ETRI; achieved 76.92% accuracy and 73.37% F1 via window size optimization.
- **SMA-Based Self-Cooling Wearable Soft Suit**:
  Contributed to the design of an SMA-based self-cooling system for wearable robotics; improved cooling efficiency through a gravity-driven mechanism and supported prototype development.

Projects

- **GBM Survival Prediction via Cellular Neighborhood Features**: [GitHub]:
  Engineered CN features from CyTOF mass cytometry data using k-NN graphs in protein expression space; achieved C-index 0.7005 (+16.4% vs. baseline) with ElasticNet Cox regression under LOO-CV.
- **SENTRY: Set-Based Hate Speech Detection**: [GitHub]:
  Developed a set-based model using DistilBERT and permutation-equivariant architecture; achieved 0.847 macro-F1, outperforming single-instance baselines.
- **LoRA vs. DAPT for Domain Adaptation in Legal QA**: [GitHub]:
  Compared LoRA and DAPT on RoBERTa using CaseHOLD; DAPT-1000 achieved 74.73% accuracy (+0.69% vs. full fine-tuning), while LoRA matched 95% performance with only 0.24% parameters.
- **Respiratory Disease Classification from Breathing Sounds**:
  Built a CNN-based model using MFCC features for 6-class respiratory sound classification; analyzed the impact of data augmentation under class imbalance, identifying overfitting as a key challenge in small-scale audio datasets.

Skills

Programming	Python, C/C++, R
Machine Learning	Deep Learning, NLP, Multimodal Learning, Transfer Learning
Frameworks & Tools	PyTorch, HuggingFace Transformers, Scikit-learn, NumPy, Pandas, Git
Languages	English(Fluent), Korean(Native)

Leadership Activities

Korean Student Association, University of North Carolina at Chapel Hill *Chapel Hill, NC, United States*

Event Management Chair May. 2026 – Present

- Planned and organized networking events including orientation and welcome gatherings for incoming students.
- Coordinated a joint sports event with Duke University and NC State University, managing cross-institution collaboration.

Department of Mechanical Systems Engineering, Sookmyung Women's University *Seoul, South Korea*

President Mar. 2019 – Dec. 2021

- Led a student council of 25+ members for three consecutive years; organized large-scale academic and social events to increase student engagement.
- Initiated a multi-department project competition (now annual) and managed a $7,000 budget with audited financial accountability.

Sookpoong Club (Traditional Korean Percussion Group) *Seoul, South Korea*

President Mar. 2019 – Aug. 2021

- Led 35+ members and organized regular performances and training camps to promote traditional Korean music and strengthen team collaboration.